Panorama Stitching & Homography Quiz

Q1. The "Pure Rotation" assumption is critical for panorama stitching because a homography matrix fundamentally assumes:

A) The camera has zero focal length.

B) All objects in the scene are at an infinite distance from the camera.

C) The camera rotates strictly about its optical center, eliminating depth-dependent shifts. ✓

D) The lens has no physical aperture.

Correct Answer: C

Q2. A standard Homography matrix is a 3×3 matrix. How many degrees of freedom does it actually possess for modeling geometric relationships?

A) 9, because there are 9 entries in the matrix.

B) 8, because it is a homogeneous matrix where one element is used for scale. ✓

C) 4, because that is the minimum number of points needed to solve it.

D) 6, to account for rotation and translation in 3D space.

Correct Answer: B

Q3. If a feature is described as being "Scale Invariant" in the SIFT algorithm, what does this actually mean for the stitching process?

A) The feature will always be the same size regardless of the camera's resolution.

B) The feature will only be detected if the camera is exactly 5 meters away from the object.

C) The algorithm can recognize the same physical point even if one photo is zoomed in or taken from a different distance than the other. ✓

D) The feature ignores the edges of the image to focus only on the center.

Correct Answer: C

Q4. In the SIFT algorithm, how is the Difference of Gaussian (DoG) function mathematically defined to identify potential features at various scales?

A) $D(x, y, \sigma) = \nabla^2 L(x, y, \sigma)$

B) $D(x, y, \sigma) = L(x, y, \sigma^2) - L(x, y, \sigma)$

C) $D(x, y, \sigma) = L(x, y, k\sigma) - L(x, y, \sigma)$ ✓

D) $D(x, y, \sigma) = G(x, y, k\sigma) / G(x, y, \sigma)$

Correct Answer: C

Q5. In SIFT's scale-space extrema detection, why is it conceptually necessary to compare a candidate pixel to 18 neighbors in adjacent scales (above and below) rather than just the 8 spatial neighbors at its own scale?

- A) To account for the 8 degrees of freedom required by the homography matrix.
- B) To ensure the feature is localized accurately in terms of illumination invariance.
- **C) To ensure the feature is a local maximum or minimum across zoom levels, making it repeatable even if the camera moves closer or farther away. ✓**
- D) To reduce the computational complexity of the K-d tree search from $O(n^2)$ to $O(n \log n)$.

Correct Answer: C

Q6. Why is RANSAC essentially required for the homography estimation step in panorama stitching?

- A) It speeds up the K-d tree feature matching process.
- B) It calculates the average position of all detected keypoints to find the center.
- **C) A single "outlier" (incorrect match) can completely corrupt a standard mathematical alignment. ✓**
- D) It is the only way to mathematically solve a 3×3 matrix.

Correct Answer: C

Q7. A photographer attempts to stitch two aerial photos of a city taken from a moving drone. Despite high-quality feature matches, the skyscrapers in the final panorama appear "ghosted" or shifted relative to the street level. What is the most likely theoretical cause?

- A) The drone moved faster than the SIFT algorithm could process orientation gradients.
- **B) The drone's translation introduced parallax, which a homography cannot model because it assumes the camera only rotates about its optical center. ✓**
- C) The skyscrapers were too tall for the Difference of Gaussian (DoG) filter to detect extrema.
- D) The radial distortion of the drone's lens ($\kappa = \pm 0.5$) made feature matching impossible.

Correct Answer: B

Q8. If feature matching identifies 100 pairs of points, but 5 of those pairs are completely wrong (e.g., a tree leaf matched to a car tire), why can't we just use the average of all 100 points to align the images?

- A) Averaging is mathematically impossible with a 3×3 matrix.
- **B) A single "outlier" (incorrect match) can completely corrupt the mathematical alignment, resulting in a distorted image. ✓**
- C) The SIFT algorithm only allows the use of 4 points at a time.
- D) Averaging would make the final panorama too blurry to see.

Correct Answer: B

Q9. Which image blending technique is specifically noted for smoothing brightness differences while keeping sharp details clear?

- A) Linear Blending
- B) Feathering
- **C) Multi-band Blending ✓**
- D) Alpha Compositing

Correct Answer: C

Q10. When a camera translates rather than just rotates, what specific error occurs that a homography cannot mathematically model?

- A) Radial distortion
- **B) Parallax ✓**
- C) Ghosting
- D) Motion blur

Correct Answer: B
